COMPUTATIONAL MATHEMATICS

Complete Theory and Revision Notes

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TOPIC 1: NUMBER SYSTEMS

1.1 Definition

A number system is a method of representing numbers using a set of digits and rules.

Common number systems in computing are:

Number system	Base	Digits used
Decimal	10	0–9
Binary	2	0, 1

.Octal	8	0–7
Hexadecimal	16	0–9, A–F

Hexadecimal letters represent these values: A = 10, B = 11, C = 12, D = 13, E = 14 and F = 15.

1.2 Decimal number system

The decimal system is the number system used in everyday calculations.

Example:

[
472=4(10^2)+7(10^1)+2(10^0)
]

[
=400+70+2
]

1.3 Binary number system

Binary uses only 0 and 1. Computers use binary to represent data and instructions.

Example:

[
1011_2=1(2^3)+0(2^2)+1(2^1)+1(2^0)
]

[
=8+0+2+1=\boxed{11_{10}}
]

1.4 Convert decimal to binary

Divide the decimal number repeatedly by 2 and record the remainders. Read the remainders from bottom to top.

Example: Convert (13_{10}) to binary.

Division	Quotient	Remainder
13 ÷ 2	6	1
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

Read upward:

Whatsapp:0720349953

$$
\boxed{13_{10}=1101_2}
$$

1.5 Convert binary to decimal

Example: Convert (11010_2) to decimal.

$$
1(2^4)+1(2^3)+0(2^2)+1(2^1)+0(2^0)
$$

$$
=16+8+0+2+0=\boxed{26_{10}}
$$

1.6 Convert decimal to hexadecimal

Divide repeatedly by 16 and read the remainders from bottom to top.

Example: Convert (254_{10}) to hexadecimal.

- (254\div16=15) remainder 14 (E)
- (15\div16=0) remainder 15 (F)

Therefore:

$$
\boxed{254_{10}=FE_{16}}
$$

1.7 Convert binary to hexadecimal

Group binary digits into groups of four, starting from the right. Add leading zeros if needed.

Example:

$$
11011110_2=1101\ 1110
$$

- (1101_2=13=D)
- (1110_2=14=E)

Therefore:

[
\boxed{11011110_2=DE_{16}}
]

Practice questions

1. Convert (25_{10}) to binary.
2. Convert (101101_2) to decimal.
3. Convert (156_{10}) to hexadecimal.
4. Convert (11110000_2) to hexadecimal.

TOPIC 2: BINARY ARITHMETIC

2.1 Binary addition

Rules:

Operation	Result
0 + 0	0
0 + 1	1
1 + 0	1
1 + 1	10
1 + 1 + 1	11

Example:

```
   1011
 + 0110
 -------
  10001
```

Therefore:

[
1011_2+0110_2=10001_2
]

Check in decimal: (11+6=17).

2.2 Binary subtraction

Rules:

- (0-0=0)
- (1-0=1)
- (1-1=0)

- (0-1) requires borrowing.

Example:

```
  1101
- 0101
-------
  1000
```

Check: (13-5=8).

2.3 Binary multiplication

The basic rules are:

- (0\times0=0)
- (0\times1=0)
- (1\times0=0)
- (1\times1=1)

Example:

```
    101
×   11
-----
    101
   1010
-----
   1111
```

Therefore:

[
101_2\times11_2=1111_2
]

2.4 Two's complement

Two's complement is a common way computers represent signed integers.

To find the two's complement of a binary number:

1. Invert every bit.
2. Add 1.

Example: Find the two's complement of `0101` using four bits.

Invert:

```
0101 → 1010
```

Add 1:

```
  1010
+ 0001
------
  1011
```

The result is `1011`. In four-bit signed two's-complement representation, this represents (-5).

Important: Four-bit two's-complement values range from (-8) to (+7).

Practice questions

1. Add (1010_2+1101_2).
2. Subtract (1100_2-0101_2).
3. Multiply (110_2\times101_2).
4. Find the four-bit two's complement of `0011`.

TOPIC 3: BOOLEAN ALGEBRA AND LOGIC GATES

3.1 Boolean algebra

Boolean algebra is a mathematical system in which variables have only two values:

- 0 = False
- 1 = True

It is important in digital electronics, computer hardware and logical decision-making.

3.2 Basic logical operations

AND operation

The output is 1 only when both inputs are 1.

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

OR operation

The output is 1 when at least one input is 1.

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

NOT operation

NOT reverses the input.

A	NOT A
0	1
1	0

3.3 Other logic gates

- **NAND:** opposite of AND.
- **NOR:** opposite of OR.
- **XOR:** output is 1 when inputs differ.
- **XNOR:** output is 1 when inputs are the same.

3.4 Boolean laws

Law	Expression
Identity	(A+0=A)
OR domination	(A+1=1)
AND identity	(A\cdot1=A)
AND domination	(A\cdot0=0)
Idempotent	(A+A=A)
Complement	(A+\overline A=1)
Double negation	(\overline{\overline A}=A)
De Morgan's law	(\overline{A+B}=\overline A\cdot\overline B)

In Boolean algebra, (+) means OR, multiplication means AND, and the bar means NOT.

Practice questions

1. Construct the truth table for (A\ AND\ NOT\ B).
2. State De Morgan's two laws.
3. Differentiate between AND and OR gates.
4. Simplify (A+0) and (A\cdot1).

TOPIC 4: SETS AND SET OPERATIONS

4.1 Definition of a set

A set is a well-defined collection of distinct objects.

Example:

[
A={1,2,3,4,5}
]

The objects in a set are called elements.

4.2 Types of sets

- **Empty set:** contains no elements, written (\varnothing).
- **Finite set:** has a limited number of elements.
- **Infinite set:** has infinitely many elements.
- **Universal set:** contains all elements being considered.
- **Subset:** every element of one set belongs to another set.

4.3 Set operations

Let:

[
A={1,2,3,4}
]

[
B={3,4,5,6}
]

Union

Union contains all elements in either set.

[
A\cup B={1,2,3,4,5,6}
]

Intersection

Intersection contains elements common to both sets.

[
A\cap B={3,4}
]

Difference

Elements in A but not in B:

[
A-B={1,2}
]

Complement

The complement of A contains elements in the universal set that are not in A.

If (U={1,2,3,4,5,6,7}), then:

[
A'={5,6,7}
]

4.4 Number of elements

The number of elements in a set is written as (n(A)).

For two finite sets:

[
n(A\cup B)=n(A)+n(B)-n(A\cap B)
]

Example: 30 students study Mathematics, 20 study Computing, and 10 study both.

[
n(A\cup B)=30+20-10=\boxed{40}
]

Therefore, 40 students study at least one of the two subjects.

Practice questions

1. Given (A={2,4,6,8}) and (B={4,8,10}), find (A\cup B).
2. Find (A\cap B).
3. If (n(A)=25), (n(B)=18) and (n(A\cap B)=8), calculate (n(A\cup B)).

TOPIC 5: ALGEBRA AND EQUATIONS

5.1 Algebraic expressions

An algebraic expression combines numbers, variables and operations.

Example:

$$
3x+5
$$

Here, (x) is the variable, 3 is its coefficient, and 5 is a constant.

5.2 Simplifying expressions

Example:

$$
3x+2x+4=5x+4
$$

Only like terms can be combined.

5.3 Linear equations

A linear equation has a variable whose highest power is 1.

Example:

$$
3x+5=20
$$

Subtract 5 from both sides:

$$
3x=15
$$

Divide by 3:

$$
\boxed{x=5}
$$

5.4 Simultaneous equations

Example:

$$
x+y=10
$$

$$
x-y=2
$$

Add the equations:

$$
2x=12\Rightarrow x=6
$$

Substitute into (x+y=10):

$$
6+y=10\Rightarrow y=4
$$

Answer:

$$
\boxed{x=6,\quad y=4}
$$

5.5 Quadratic equations

A quadratic equation has the general form:

$$
ax^2+bx+c=0,\quad a\ne0
$$

It can be solved by factorization, completing the square or the quadratic formula.

The quadratic formula is:

$$
x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}
$$

Example:

$$
x^2-5x+6=0
$$

Factorize:

$$
(x-2)(x-3)=0
$$

Therefore:

$$
\boxed{x=2\text{ or }x=3}
$$

5.6 Percentages

$$
\text{Percentage}=\frac{\text{Part}}{\text{Whole}}\times100\%
$$

Example: Find 15% of KSh 2,000.

$$
\frac{15}{100}\times2000=\boxed{\text{KSh }300}
$$

5.7 Ratio and proportion

A ratio compares quantities.

Example:

The ratio of boys to girls is (2:3). If there are 25 students, the total number of parts is 5.

One part:

$$
25\div5=5
$$

Boys:

$$
2\times5=10
$$

Girls:

$$
3\times5=15
$$

Practice questions

1. Solve (5x-7=18).
2. Solve (2x+y=11) and (x-y=1).
3. Solve (x^2-7x+12=0).
4. Calculate 12% of KSh 5,000.
5. Divide 72 in the ratio (5:3).

TOPIC 6: INDICES, LOGARITHMS AND SURDS

6.1 Indices

An index or exponent shows how many times a number is multiplied by itself.

[
2^3=2\times2\times2=8
]

6.2 Laws of indices

For suitable non-zero values:

[
a^m\times a^n=a^{m+n}
]

[
\frac{a^m}{a^n}=a^{m-n}
]

[
(a^m)^n=a^{mn}
]

[
a^0=1
]

[
a^{-n}=\frac{1}{a^n}
]

[
a^{1/n}=\sqrt[n]{a}
]

Example:

[
2^3\times2^4=2^7=\boxed{128}
]

6.3 Logarithms

A logarithm answers the question: to what power must a base be raised to produce a number?

[
\log_b x=y\iff b^y=x
]

Example:

[
\log_2 8=3
]

because:

[
2^3=8
]

6.4 Laws of logarithms

For positive (M,N) and a valid base (b):

[
\log_b(MN)=\log_bM+\log_bN
]

[
\log_b\left(\frac MN\right)=\log_bM-\log_bN
]

[
\log_b(M^n)=n\log_bM
]

6.5 Surds

A surd is an irrational root expressed in radical form.

Example:

[
\sqrt{12}=\sqrt{4\times3}=2\sqrt3
]

Another example:

[
\sqrt{50}=5\sqrt2
]

Practice questions

1. Simplify (3^2\times3^4).
2. Evaluate (16^{1/2}).
3. Find (\log_{10}1000).
4. Simplify (\sqrt{72}).

TOPIC 7: MATRICES AND DETERMINANTS

7.1 Definition of a matrix

A matrix is a rectangular arrangement of numbers in rows and columns.

Example:

[
A=\begin{bmatrix}1&2\3&4\end{bmatrix}
]

This is a (2\times2) matrix.

7.2 Types of matrices

- Row matrix: one row.
- Column matrix: one column.
- Square matrix: equal numbers of rows and columns.
- Zero matrix: all entries are zero.
- Identity matrix: diagonal entries are 1 and other entries are 0.

7.3 Matrix addition

Matrices must have the same dimensions.

[
A=\begin{bmatrix}1&2\3&4\end{bmatrix},
\quad
B=\begin{bmatrix}5&6\7&8\end{bmatrix}
]

[
A+B=\begin{bmatrix}6&8\10&12\end{bmatrix}
]

7.4 Matrix subtraction

[
A-B=\begin{bmatrix}-4&-4\-4&-4\end{bmatrix}
]

7.5 Matrix multiplication

For two (2\times2) matrices:

[
AB=\begin{bmatrix}
a&b\c&d
\end{bmatrix}
\begin{bmatrix}
e&f\g&h
\end{bmatrix}
]

[
=\begin{bmatrix}
ae+bg&af+bh\
ce+dg&cf+dh
\end{bmatrix}
]

The number of columns in the first matrix must equal the number of rows in the second matrix.

7.6 Determinant of a (2\times2) matrix

For:

[
A=\begin{bmatrix}a&b\c&d\end{bmatrix}
]

The determinant is:

[
|A|=ad-bc
]

Example:

[
A=\begin{bmatrix}2&3\1&4\end{bmatrix}
]

[
|A|=(2)(4)-(3)(1)=8-3=\boxed5
]

7.7 Inverse of a (2\times2) matrix

If the determinant is not zero:

[
A^{-1}=\frac{1}{ad-bc}
\begin{bmatrix}d&-b\-c&a\end{bmatrix}
]

Practice questions

1. Add (\begin{bmatrix}2&3\1&4\end{bmatrix}) and (\begin{bmatrix}1&5\2&0\end{bmatrix}).
2. Find the determinant of (\begin{bmatrix}4&2\3&5\end{bmatrix}).
3. State the condition required for a square matrix to have an inverse.

TOPIC 8: SEQUENCES AND SERIES

8.1 Sequence

A sequence is an ordered list of numbers following a rule.

Example:

[
2,4,6,8,10,\ldots
]

8.2 Arithmetic progression (AP)

An arithmetic progression has a constant difference between consecutive terms.

Example:

[
3,7,11,15,\ldots
]

Common difference:

[
d=7-3=4
]

nth term

[
a_n=a+(n-1)d
]

where:

- (a) = first term
- (d) = common difference
- (n) = term number

Example: Find the 10th term of (3,7,11,\ldots).

[
a_{10}=3+(10-1)4
]

[
=3+36=\boxed{39}
]

Sum of an AP

[
S_n=\frac n2[2a+(n-1)d]
]

Alternatively:

[
S_n=\frac n2(a+l)
]

where (l) is the last term.

8.3 Geometric progression (GP)

A geometric progression has a constant ratio between consecutive terms.

Example:

[
2,6,18,54,\ldots
]

Common ratio:

[
r=\frac62=3
]

nth term

[
a_n=ar^{n-1}
]

Sum of a finite GP

For (r\ne1):

[
S_n=\frac{a(r^n-1)}{r-1}
]

Practice questions

1. Find the 15th term of (2,5,8,11,\ldots).
2. Find the sum of the first 10 terms of (1,3,5,7,\ldots).
3. Find the 6th term of (3,6,12,24,\ldots).

TOPIC 9: FUNCTIONS AND GRAPHS

9.1 Function

A function is a rule that assigns exactly one output to each permitted input.

Example:

$$
f(x)=2x+3
$$

If $x=4$:

$$
f(4)=2(4)+3=\boxed{11}
$$

9.2 Linear functions

A linear function has the form:

$$
y=mx+c
$$

where:

- m = gradient or slope
- c = y-intercept

Example:

$$
y=2x+1
$$

When $x=0$, $y=1$. This gives the y-intercept.

9.3 Gradient

For two points (x_1,y_1) and (x_2,y_2):

$$
m=\frac{y_2-y_1}{x_2-x_1}
$$

Example: Find the gradient between $(1,2)$ and $(3,8)$.

$$
m=\frac{8-2}{3-1}=\frac62=\boxed3
$$

9.4 Quadratic functions

A quadratic function has the form:

[
y=ax^2+bx+c
]

Its graph is a parabola.

- If (a>0), it opens upward.
- If (a<0), it opens downward.

9.5 Applications of functions

Functions are used to represent relationships in:

- Business calculations
- Computer algorithms
- Cost and revenue analysis
- Scientific models
- Engineering calculations

Practice questions

1. If (f(x)=3x-2), find (f(5)).
2. Find the gradient between ((2,4)) and ((6,12)).
3. State the general form of a linear function.

TOPIC 10: STATISTICS

10.1 Definition

Statistics is the branch of mathematics concerned with collecting, organizing, presenting, analyzing and interpreting data.

10.2 Types of data

Qualitative data

Describes categories or characteristics.

Examples: course, colour, county.

Quantitative data

Expressed numerically.

Examples: age, marks, height.

Quantitative data can be discrete or continuous.

10.3 Mean

The arithmetic mean is calculated by dividing the total of all observations by the number of observations.

$$
\bar{x}=\frac{\sum x}{n}
$$

Example:

Data: (4,6,8,10,12)

$$
\bar{x}=\frac{4+6+8+10+12}{5}
=\frac{40}{5}=\boxed{8}
$$

10.4 Median

The median is the middle value after arranging data in ascending order.

Example:

(3,5,7,9,11)

Median = 7.

For an even number of observations, take the average of the two middle values.

Example:

(2,4,6,8)

$$
\text{Median}=\frac{4+6}{2}=\boxed{5}
$$

10.5 Mode

The mode is the value that occurs most frequently.

Example:

(2,3,3,4,5)

Mode = 3.

10.6 Range

$$
\text{Range}=\text{Maximum}-\text{Minimum}
$$

For (4,7,9,12,15):

$$
15-4=\boxed{11}
$$

10.7 Variance and standard deviation

Variance measures the spread of observations around their mean.

For a population:

$$
\sigma^2=\frac{\sum(x-\mu)^2}{N}
$$

Population standard deviation:

$$
\sigma=\sqrt{\sigma^2}
$$

For a sample, the usual sample variance formula divides by (n-1), not (n).

10.8 Data presentation

Common methods include:

- Tables
- Bar charts
- Pie charts
- Histograms
- Line graphs
- Frequency distributions

Practice questions

For the data (5,7,7,9,12), calculate:

1. Mean
2. Median
3. Mode
4. Range

TOPIC 11: PROBABILITY

11.1 Definition

Probability measures the likelihood that an event will occur.

For equally likely outcomes:

[
P(E)=\frac{\text{Number of favourable outcomes}}{\text{Total possible outcomes}}
]

Probability lies between 0 and 1.

- 0 means impossible.
- 1 means certain.

11.2 Example: Tossing a coin

A fair coin has two possible outcomes: Head and Tail.

[
P(\text{Head})=\frac12
]

11.3 Example: Rolling a die

A fair die has six possible outcomes.

Probability of rolling a 4:

[
P(4)=\frac16
]

Probability of rolling an even number:

$$
P(\text{even})=\frac36=\boxed{\frac12}
$$

11.4 Complementary events

$$
P(E')=1-P(E)
$$

If the probability of rain is 0.3:

$$
P(\text{no rain})=1-0.3=\boxed{0.7}
$$

11.5 Addition rule

For two events:

$$
P(A\cup B)=P(A)+P(B)-P(A\cap B)
$$

If events are mutually exclusive, their intersection is zero.

11.6 Independent events

For independent events:

$$
P(A\cap B)=P(A)\times P(B)
$$

Example: Probability of getting two heads in two independent tosses of a fair coin:

$$
\frac12\times\frac12=\boxed{\frac14}
$$

Practice questions

1. Find the probability of rolling a 6 on a fair die.
2. Find the probability of getting an odd number on a fair die.
3. A bag contains 3 red balls and 7 blue balls. Find the probability of selecting a red ball at random.
4. Find the probability of not selecting a red ball from that bag.

TOPIC 12: NUMERICAL METHODS

Numerical methods use arithmetic procedures to find approximate solutions to mathematical problems.

They are useful when an exact solution is difficult or unavailable.

12.1 Bisection method

The bisection method estimates a root of an equation by repeatedly halving an interval in which a continuous function changes sign.

For an interval ([a,b]), calculate:

[
c=\frac{a+b}{2}
]

Then choose the half-interval where the sign change remains.

Example: Approximate a root of:

[
f(x)=x^2-4
]

The positive root lies in ([1,3]) because:

[
f(1)=-3,\qquad f(3)=5
]

First midpoint:

[
c=\frac{1+3}{2}=2
]

Since (f(2)=0), the root is exactly:

[
\boxed{x=2}
]

12.2 Newton-Raphson method

This method uses a function and its derivative to approximate a root.

Formula:

$$
x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}
$$

It requires a suitable initial estimate and a non-zero derivative at the iterations used.

12.3 Trapezoidal rule

The trapezoidal rule estimates the area under a curve using trapezoids.

For one interval:

$$
\int_a^b f(x),dx\approx\frac{b-a}{2}[f(a)+f(b)]
$$

For (n) equal intervals with width (h):

$$
\int_a^b f(x),dx\approx
\frac h2[y_0+2(y_1+\cdots+y_{n-1})+y_n]
$$

12.4 Applications of numerical methods

- Engineering
- Scientific computing
- Financial modelling
- Simulation
- Data analysis
- Computer graphics

Practice questions

1. Explain the purpose of numerical methods.
2. State the bisection midpoint formula.
3. Write the Newton-Raphson formula.
4. State one limitation of the bisection method and one limitation of Newton-Raphson.

TOPIC 13: ERRORS, APPROXIMATIONS AND COMPUTER APPLICATIONS

13.1 Rounding off

Rounding reduces a number to a specified number of decimal places or significant figures.

Example: Round 3.146 to two decimal places.

The third decimal digit is 6, so round up:

[
\boxed{3.15}
]

13.2 Significant figures

Significant figures are the meaningful digits used to express the precision of a number.

Example:

[
0.004572
]

To three significant figures:

[
\boxed{0.00457}
]

13.3 Absolute error

Absolute error is the magnitude of the difference between an approximate value and the true value.

[
E_a=|\text{True value}-\text{Approximate value}|
]

Example: True value = 10, approximate value = 9.8.

$$
E_a=|10-9.8|=\boxed{0.2}
$$

13.4 Relative error

$$
E_r=\frac{E_a}{|\text{True value}|}
$$

Using the example above:

$$
E_r=\frac{0.2}{10}=\boxed{0.02}
$$

13.5 Percentage error

$$
E_p=E_r\times100\%
$$

$$
E_p=0.02\times100\%=\boxed{2\%}
$$

13.6 Applications of computational mathematics

Computational mathematics is used in:

- Programming algorithms
- Data analysis
- Financial calculations
- Scientific simulations
- Engineering
- Computer graphics
- Artificial intelligence
- Numerical problem-solving

13.7 Limitations of computational calculations

- Rounding can introduce errors.
- Incorrect input can produce incorrect results.
- Numerical methods may only provide approximations.
- Some calculations may accumulate floating-point errors.
- An unsuitable algorithm can produce inaccurate or inefficient results.

TOPIC 14: REVISION QUESTIONS AND WORKED ANSWERS

Section A: Multiple-choice questions

1. What is the base of the binary number system?

A. 2
B. 8
C. 10
D. 16

Answer: A

2. Convert (1010_2) to decimal.

A. 8
B. 10
C. 12
D. 14

Answer: B

3. Which logic gate gives output 1 only when both inputs are 1?

A. OR
B. NOT
C. AND
D. XOR

Answer: C

4. What is the median of (2,4,6,8,10)?

A. 4
B. 5
C. 6
D. 8

Answer: C

5. What is the probability of getting a head when tossing a fair coin?

A. 0
B. (1/4)
C. (1/2)
D. 1

Answer: C

6. What is the formula for the nth term of an arithmetic progression?

A. (a_n=ar^{n-1})
B. (a_n=a+(n-1)d)
C. (a_n=a+d^n)
D. (a_n=nd)

Answer: B

7. Which formula calculates the determinant of a (2\times2) matrix?

A. (ad+bc)
B. (ab-cd)
C. (ad-bc)
D. (a+b+c+d)

Answer: C

8. What is the absolute error if the true value is 50 and the approximation is 49?

A. 0
B. 1
C. 2
D. 5

Answer: B

Section B: Short-answer questions

1. Define computational mathematics.
2. State four number systems used in computing.
3. Convert (11001_2) to decimal.
4. Convert (35_{10}) to binary.
5. State the basic rules of binary addition.
6. Explain Boolean algebra.
7. State four logic gates.
8. Define a set.
9. Differentiate between union and intersection.
10. Solve (4x+8=28).
11. State three laws of indices.
12. Define a matrix.
13. Find the determinant of (\begin{bmatrix}3&2\1&5\end{bmatrix}).
14. Define an arithmetic progression.
15. Define mean, median and mode.
16. Define probability.
17. State the formula for absolute error.
18. Explain the bisection method.

19. State two applications of computational mathematics.
20. Differentiate between absolute error and percentage error.

Section C: Worked problems

Question 1: Number conversion

Convert (45_{10}) to binary.

Solution

```
45 ÷ 2 = 22 remainder 1
22 ÷ 2 = 11 remainder 0
11 ÷ 2 = 5  remainder 1
5  ÷ 2 = 2  remainder 1
2  ÷ 2 = 1  remainder 0
1  ÷ 2 = 0  remainder 1
```

Read from bottom to top:

[
\boxed{45_{10}=101101_2}
]

Question 2: Mean

Find the mean of 12, 15, 18, 20 and 25.

Solution

[
\bar{x}=\frac{12+15+18+20+25}{5}
]

[
=\frac{90}{5}=\boxed{18}
]

Question 3: Linear equation

Solve:

[
5x-10=30
]

Add 10 to both sides:

[
5x=40
]

Divide by 5:

[
\boxed{x=8}
]

Question 4: Probability

A bag contains 4 green balls and 6 yellow balls. Find the probability of selecting a green ball.

Solution

Total balls:

[
4+6=10
]

[
P(\text{green})=\frac4{10}=\boxed{\frac25}
]

Question 5: Matrix determinant

Find the determinant:

[
A=\begin{bmatrix}5&2\3&4\end{bmatrix}
]

Solution

[
|A|=(5)(4)-(2)(3)
]

[
=20-6=\boxed{14}
]

Question 6: Percentage error

A measured length is 19.6 cm while the true length is 20 cm. Calculate the percentage error.

Solution

Absolute error:

$$
E_a=|20-19.6|=0.4
$$

Relative error:

$$
E_r=\frac{0.4}{20}=0.02
$$

Percentage error:

$$
E_p=0.02\times100\%=\boxed{2\%}
$$

FINAL REVISION CHECKLIST

Before your examination, make sure you can:

- ☐ Convert decimal, binary and hexadecimal numbers.
- ☐ Perform binary addition, subtraction and multiplication.
- ☐ Apply two's-complement representation.
- ☐ Construct truth tables.
- ☐ Apply Boolean algebra laws.
- ☐ Calculate set unions and intersections.

- 

  Solve linear and quadratic equations.

- 

  Apply laws of indices and logarithms.

- 

  Perform matrix operations.

- 

  Calculate determinants and inverses of suitable matrices.

- 

  Solve arithmetic and geometric progression problems.

- 

  Evaluate functions and calculate gradients.

- 

  Calculate mean, median, mode and range.

- 

  Solve elementary probability problems.

- 

  Explain basic numerical methods.

- 

  Calculate absolute, relative and percentage errors.

- 

  Round numbers to decimal places and significant figures.

End of Computational Mathematics notes.

Note: Computational Mathematics syllabuses vary between TVET courses and examination levels. These notes cover common foundational topics; use your official course syllabus or module outline to confirm the exact examinable topics.